

Problem

A transistor has the following parameters in a common-emitter circuit:

$$\begin{array}{l} h_{ie} = 2,640 \, \Omega, \quad h_{re} = 2.6 \times 10^{-4} \\ h_{fe} = 72, \quad h_{oe} = 16 \, \mu\text{S}, \quad R_L = 100 \, \text{k}\Omega \end{array}$$

What is the voltage amplification of the transistor? How many decibels gain is this?

Step-by-step solution

Step 1 of 4

Consider the following common-emitter circuit data:

Base input impedance, $h_{ie} = 2640 \, \Omega$

Reverse voltage feedback ratio, $h_{fe} = 72$

Base-collector current gain, $h_{re} = 2.6 \times 10^{-4}$

Output admittance, $h_{oe} = 16 \, \mu\text{S}$

Load resistance, $R_L = 100 \, \text{k}\Omega$

Step 2 of 4

Calculate the voltage amplification of the transistor.

$$\begin{aligned} A_v &= \frac{-h_{fe} R_L}{h_{ie} + (h_{ie} h_{oe} - h_{re} h_{fe}) R_L} \\ &= \frac{-(72)(100 \times 10^3)}{2640 + [(2640)(16 \times 10^{-6}) - (2.6 \times 10^{-4})(72)](100 \times 10^3)} \\ &= \frac{-72 \times 10^5}{2640 + [0.04224 - 0.01872](100 \times 10^3)} \\ &= \frac{-72 \times 10^5}{2640 + (0.02352)(100 \times 10^3)} \\ &= \frac{-72 \times 10^5}{2640 + 2352} \\ &= \frac{-72 \times 10^5}{4992} \\ &= -1442.3 \end{aligned}$$

Step 3 of 4

Therefore, the voltage amplification of the transistor, A_v is -1442.3.

Step 4 of 4

Calculate the dc gain of the transistor circuit.

$$\begin{aligned}\text{dc gain} &= 20 \log |A_v| \\ &= 20 \log |1442.3| \\ &= 20(3.159) \\ &= 63.18 \text{ dB}\end{aligned}$$

Therefore, dc gain of the transistor circuit is 63.18 dB.